C Programming Assignment Problems

Topic 1: Strings (10 Problems)

1. String Length without `strlen :`
   Write a function that computes the length of a string without using the standard library function `strlen`.
2. Palindrome Check :
   Check whether a given string is a palindrome (reads same forwards and backwards), ignoring case and non-alphanumeric characters.
3. Reverse a String :
   Reverse a string in place without using an extra array.
4. Count Vowels and Consonants :
   Count the number of vowels and consonants in a string.
5. Remove Duplicate Characters :
   Remove all duplicate characters from a string, keeping only the first occurrence of each character.
6. String Concatenation without `strcat :`
   Concatenate two strings without using `strcat`.
7. Find Substring :
   Find all occurrences of a substring in a given string and print their starting indices.
8. Replace Character :
   Replace all occurrences of a given character in a string with another character.
9. Word Count :
   Count the number of words in a string (words separated by spaces, tabs, or newlines).
10. String to Integer (`atoi` implementation) :
    Convert a numeric string (possibly with leading signs and spaces) to an integer without using `atoi`.

Topic 2: Recursion (9 Problems)

1. Factorial :
   Compute the factorial of a non-negative integer using recursion.
2. Fibonacci Series :
   Print the first n Fibonacci numbers using recursion.
3. Sum of Digits :
   Find the sum of digits of a positive integer recursively.
4. Power Function :
   Implement `x^n` (x raised to power n) using recursion.
5. Greatest Common Divisor (GCD) :
   Compute GCD of two integers using Euclid's algorithm recursively.
6. Reverse a String Recursively :
   Reverse a string using recursion.
7. Check Palindrome (Recursive) :
   Check if a string is palindrome using recursion.
8. Print Array Elements :
   Print all elements of an array recursively (forward and backward).
9. Count Occurrences in Array :
   Given an array and a key, count how many times the key appears using recursion.

Topic 3: File Handling (6 Problems)

1. File Copy :
   Copy contents from one text file to another.
2. Character Count :
   Count the number of characters, words, and lines in a text file.
3. File to Uppercase :
   Read a text file and write its uppercase version to another file.
4. Search in File :
   Search for a given word in a file and print all line numbers where it appears.
5. Store Student Records :
   Write student data (roll, name, marks) into a binary file, then read and display.
6. Merge Two Files :
   Merge contents of two files into a third file.

Topic 4: Pointers (8 Problems)

1. Swap Two Numbers :
   Swap two integers using pointers.
2. Pointer Arithmetic :
   Traverse an array using pointer arithmetic and print elements.
3. Find Largest Element :
   Using pointers, find the largest element in an array.
4. Reverse Array using Pointers :
   Reverse an array using pointers without indexing.
5. String Length using Pointer :
   Implement `strlen` using a pointer.
6. Sum of Array using Pointers :
   Compute sum of array elements using pointers.
7. Pointer to Function :
   Create an array of function pointers to perform different arithmetic operations (add, subtract, multiply, divide).
8. Dynamic Memory Allocation :
   Allocate memory for an array of n integers using `malloc`, take input, find average, and free memory.

Topic 5: Structures (6 Problems)

1. Student Record :
   Define a structure `Student` with name, roll, marks. Input and display n students.
2. Time Structure :
   Define `Time` (hours, minutes, seconds). Write function to add two times and return result.
3. Employee Database :
   Structure `Employee` (id, name, salary). Find the highest salary employee.
4. Array of Structures
   Store book records (title, author, price). Sort by price.

5. Structure with Pointer
   Dynamically allocate an array of structures for n students.
6. Union vs Structure
   Demonstrate difference between union and structure by creating a `Data` type that can hold int, float, or char.

Topic 6: Arrays (10 Problems)

1. Find Min and Max :
   Find minimum and maximum elements in an array.
2. Reverse Array :
   Reverse the array without using extra array.
3. Second Largest :
   Find second largest element in an array.
4. Remove Duplicates from Sorted Array :
   Remove duplicates from a sorted array in place.
5. Left Rotate :
   Left rotate an array by k positions.
6. Merge Two Sorted Arrays :
   Merge two sorted arrays into a third sorted array.
7. Matrix Addition :
   Add two matrices (2D arrays).
8. Transpose of Matrix :
   Find transpose of a square matrix without extra array.
9. Bubble Sort :
   Sort an array using bubble sort.
10. Linear and Binary Search :
    Implement both linear and binary search on an array.

Topic 7: Functions (9 Problems)

1. Prime Check :
   Write a function that returns 1 if number is prime, else 0.5

2. Armstrong Number :
   Function to check if a number is Armstrong (sum of cubes of digits equals number).
3. GCD using Function :
   Write a function to compute GCD (using Euclidean algorithm, non-recursive).
4. Factorial using Function :
   Function to compute factorial (non-recursive).
5. Swap by Value vs Reference :
   Demonstrate difference between call by value and call by reference using swap functions.
6. Area and Perimeter :
   Functions to compute area and perimeter of circle, rectangle, triangle.
7. Passing Array to Function :
   Write a function to compute average of array elements.
8. Function returning Array :
   Write a function that returns a dynamically allocated array (e.g., first n Fibonacci numbers).
9. Static Variable :
   Write a function that keeps count of how many times it has been called using a static variable..